

Multilayer Formation, Interfacial Binding, and Stability of Self-Assembled Molecules in Perovskite Solar Cells

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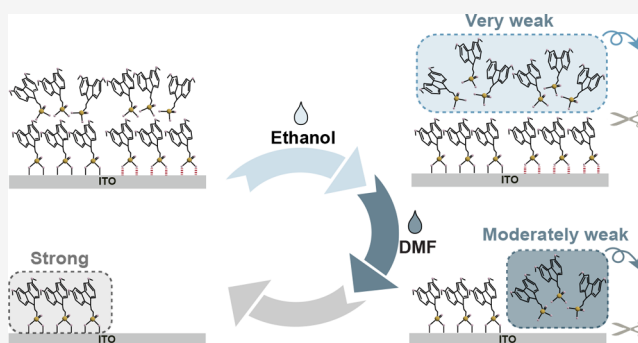


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ABSTRACT: Carbazole-based self-assembled monolayers (SAMs) have played a key role in advancing the efficiency and stability of inverted perovskite solar cells (PSCs). However, weakly bound SAM molecules can be removed by polar solvents used in perovskite deposition. The loss of SAM molecules, especially those bound to the transparent conductive oxide substrate, disrupts interfacial energetics and accelerates PSC degradation. Quantifying initial SAM coverage and tracking its loss during fabrication are therefore critical yet experimentally challenging. Here, we develop a computational–experimental approach combining density functional theory with multimodal surface characterization, including X-ray photoelectron spectroscopy and cyclic voltammetry, to selectively remove and quantify SAM molecules in distinct adsorption modes, enabling reconstruction of their initial structure and evolution during processing. We reveal that SAMs, although commonly treated as single molecular layers, comprise multiple layers. Furthermore, SAMs undergo major restructuring upon exposure to *N,N*-dimethylformamide (DMF), a common perovskite precursor solvent, which removes all upper layers and nearly half of the first-layer molecules. To mitigate these losses, we implemented a redeposition strategy introducing new SAM molecules onto the DMF-washed SAM. Redeposition resulted in 13–21% more molecules retained after a second DMF wash compared to the DMF-washed SAM without redeposition. Devices with redeposited SAMs retain 90% of the initial efficiency for 480 h under 85 °C and 50% relative humidity—a 5-fold improvement in operational stability compared to unwashed samples. More broadly, this work establishes a first-layer-sensitive, quantitative method to track SAM evolution during fabrication and offers a simple, generalizable route to durable interfaces in PSCs.



INTRODUCTION

Since their emergence in 2009,¹ perovskite solar cells (PSCs) have experienced great advancements in both efficiency² and stability.³ These performance gains have been driven by strategies such as active layer compositional engineering,^{4–6} surface passivation,^{7–10} and device architecture optimization.¹¹ Among these, the implementation of carbazole-based self-assembled molecules (SAMs), particularly [2-(9*H*-carbazol-9-yl)ethyl]phosphonic acid (2PACz) and its derivatives, as the hole transport layer (HTL) has emerged as a promising approach for improved performance in p–i–n PSCs.¹² The effectiveness of SAMs arises largely from their tunable molecular structures, which enable precise surface passivation and work function modulation, driving significant improvements in the efficiencies and stabilities of single-junction and tandem PSCs.¹³ However, even with the use of SAM HTLs, PSC efficiencies still are below that of silicon-based solar cells,² which are the industry standard, and their stability remains insufficient for commercial viability (20- to 25-year operational

lifetime¹⁴), presenting significant barriers to real-world deployment.

A major factor limiting the performance of p–i–n PSCs is the poor stability of hole-transporting SAMs on transparent conductive oxide (TCO) substrates. For reliable device operation, SAM molecules must remain strongly bound to the substrate throughout fabrication and use.^{15–18} In practice, however, SAM stability is often compromised by incomplete or unstable interfacial binding. This instability can arise from both a limited density of surface adsorption sites and weak intermolecular interactions within the SAM. Regarding the former, phosphonic acid anchoring groups present in many

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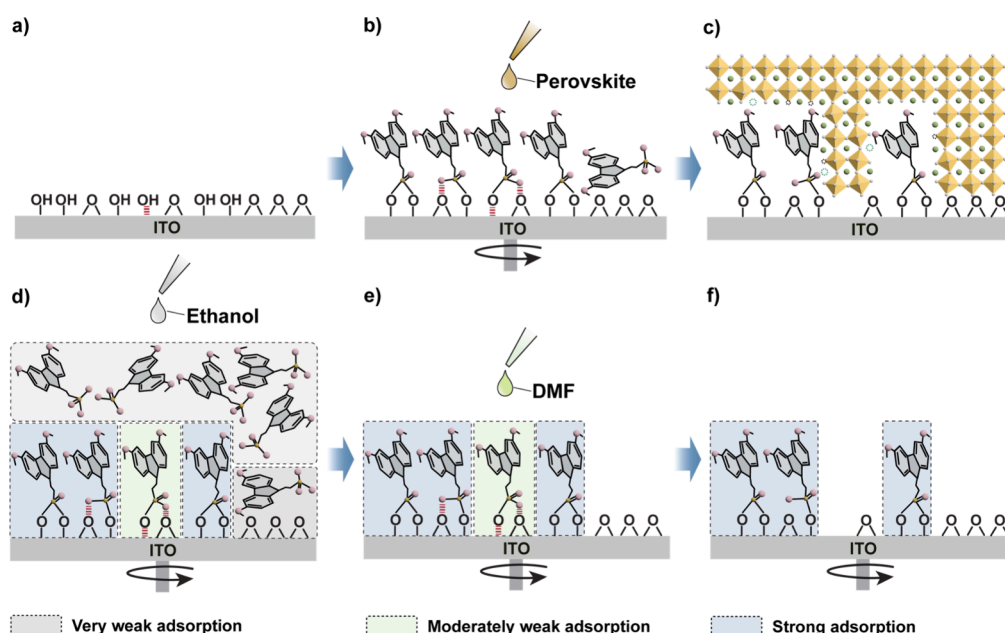


Figure 1. Illustration depicting the effect of perovskite deposition on the SAM structure and the proposed selective solvent washing strategy. (a) Plasma-treated ITO surface consisting of both hydroxylated and nonhydroxylated regions. Strong adsorption sites are indicated by solid black lines; weak adsorption sites by dashed red lines. (b) Formation of the SAM, where the distribution of adsorption modes is governed by the underlying ITO surface chemistry. The perovskite precursor solution will subsequently remove weakly bound species. (c) SAM structure after perovskite deposition. (d–f) Illustration of the selective solvent washing strategy to quantify the distribution of SAM adsorption modes: (d) ethanol washing selectively removes secondary layers and very weakly adsorbed molecules in the first layer, such as those bound through the carbazole group; (e) DMF washing removes both secondary layers and a broader range of weakly bound species in the first layer, including those adsorbed via hydrogen bonding; (f) final SAM structure after DMF washing, where only reactively adsorbed SAM molecules remain. This sequential washing approach enables reconstruction and quantification of the SAM adsorption mode distribution.

SAM molecules are known for their strong reactive adsorption onto TCO surfaces.^{19,20} This process occurs via a heterocondensation reaction, where phosphonic acid reacts with surface hydroxyl groups (–OH) on the TCO substrate, forming covalent bonds and releasing water molecules.^{21,22} However, in practice, even after surface treatments like UV-ozone or oxygen plasma, the TCO surface remains only partially hydroxylated, retaining a mixture of hydroxyl groups and metal oxide (O^{2–}) sites (Figure 1a).²³ Moreover, many of the hydroxyl groups are only weakly bound to the surface, making them less available for stable covalent bond formation.^{24,25} This incomplete and unstable hydroxylation inherently restricts the formation of strong covalent bonds with SAM molecules across the entire surface while simultaneously promoting weaker, nonreactive adsorption mechanisms at the SAM/TCO interface (Figure 1b). Regarding the latter, SAM formation, particularly for SAM molecules with flexible alkyl linkers, can yield noncompact molecular packing that weakens intermolecular stabilization, further diminishing overall SAM stability.²⁶

Recent research has highlighted the impact of weak interfacial interactions on SAM formation, PSC efficiency, and long-term stability. For example, Tang et al. and Luo et al. both demonstrated through *N,N*-dimethylformamide (DMF) solvent washing that weakly bound hydroxyl groups at the indium tin oxide (ITO) surface can be removed and that replacing them with covalently bonded hydroxyl groups, either via atomic layer deposition of an ITO surface²⁷ or solvent etching,²⁸ greatly improves SAM binding and PSC performance. This performance improvement likely stems from the fact that weakly adsorbed SAM molecules are not only vulnerable to degradation under operational stressors such as heat and

light exposure^{29,30} but can also be stripped away during PSC fabrication. Specifically, during perovskite deposition, precursor solutions containing polar solvents like DMF and dimethyl sulfoxide (DMSO) dissolve weakly bound SAM molecules,^{31,32} leading to incomplete SAM coverage and unintended contamination of the perovskite layer. As a result, even if a thin, uniform SAM initially forms on the TCO substrate, its composition may change after perovskite deposition due to the removal of weakly bound species (Figure 1c). Therefore, to fundamentally understand how the SAM composition evolves throughout PSC fabrication, it is critical to quantify the distribution of different SAM adsorption modes on the TCO surface, particularly before and after perovskite deposition.

Previous studies have qualitatively assessed the binding strength distribution of SAM molecules on TCO substrates using techniques like mechanical peeling of perovskite films to visualize SAM adhesion to the underlying TCO²⁹ and semiquantitative approaches such as X-ray photoelectron spectroscopy (XPS) to track changes in SAM coverage after solvent washing.²⁷ However, no existing work has quantitatively described the distribution of adsorption modes between the SAM and TCO substrate, both before and after perovskite deposition. This challenge is further complicated by the fact that SAMs may unintentionally adopt multilayer structures^{33,34} rather than ideal monolayers. In such cases, weakly bound molecules in secondary layers may obscure signals from the molecules within the first layer, limiting the ability of conventional techniques to selectively resolve the binding strength distribution directly at the SAM/TCO interface. Therefore, beyond simply developing a quantitative approach to measure the distribution of SAM adsorption modes on TCO surfaces, it is critical that such an approach also be first-

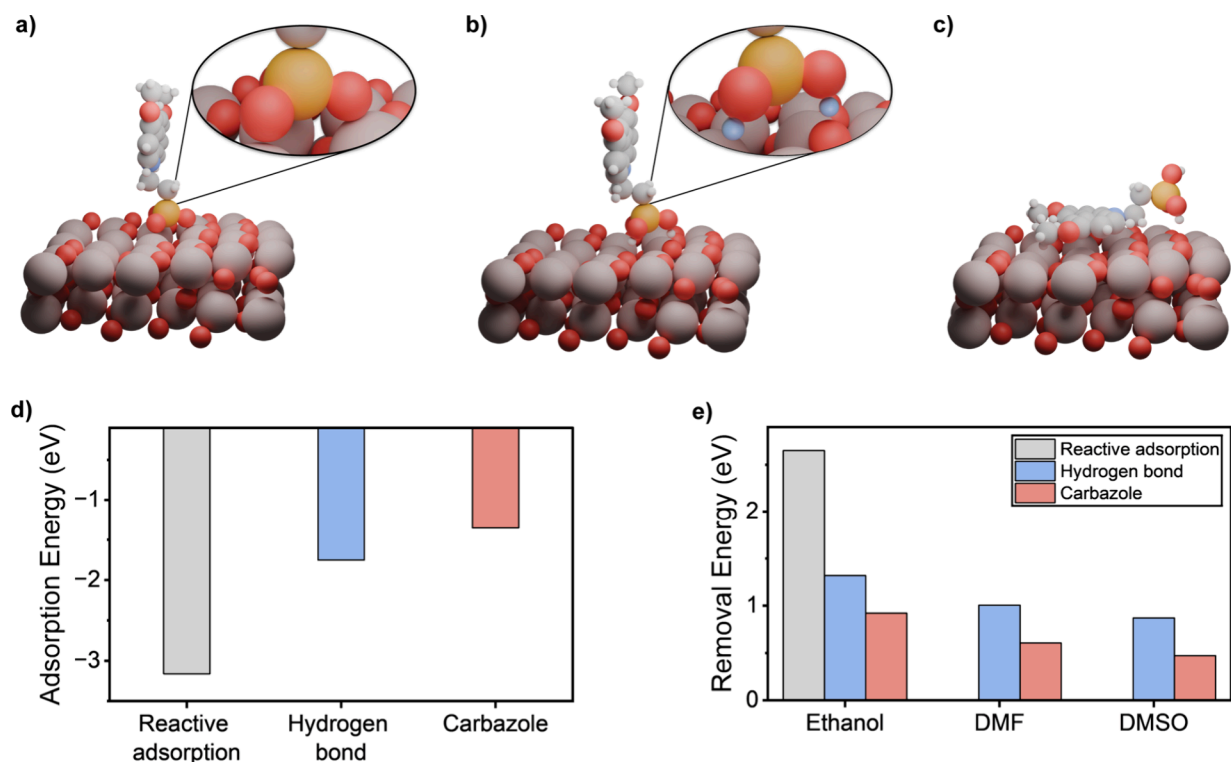


Figure 2. DFT calculations of SAM adsorption and solvent-induced removal. (a–c) DFT-optimized structures of three proposed MeO-2PACz adsorption configurations on In₂O₃(111): (a) reactive adsorption via a heterocondensation reaction; (b) nonreactive adsorption via hydrogen bonding, where the molecule is oriented perpendicular to the surface so that the phosphonic acid group could hydrogen-bond; and (c) nonreactive adsorption through the carbazole group, where the molecule is oriented parallel to the surface so that the carbazole group could interact with the surface. Insets in (a) and (b) highlight the phosphonic acid regions of MeO-2PACz to emphasize deprotonation in the reactive adsorption case and the lack thereof in the hydrogen bonding mechanism. Hydrogen atoms in the inset of (b) are highlighted in blue to more easily visualize. (d) Corresponding DFT-calculated adsorption energies for each mechanism for MeO-2PACz, shown left to right as reactive adsorption, hydrogen bonding, and carbazole-based adsorption. The reactive adsorption energy was computed following the deprotonation step and water formation to directly quantify the strength of the chemisorbed SAM. (e) DFT-calculated removal energies of adsorbed MeO-2PACz by ethanol, DMF, and DMSO. Bars represent removal energies for MeO-2PACz adsorbed via reactive adsorption (gray), hydrogen bonding (blue), and the carbazole group (red). Due to the lack of hydrogen bond sites on DMF, DMSO, and the reactively adsorbed MeO-2PACz molecules, both solvents are assumed to be ineffective at displacing reactively adsorbed MeO-2PACz.

layer-sensitive, capable of selectively probing the substrate-bound monolayer. Without such insight, efforts to improve SAM stability remain largely empirical, making it difficult to establish universal design principles or predict how the SAM/TCO interface evolves throughout PSC fabrication.

In this work, we present a combined computational–experimental approach that quantitatively differentiates between strongly and weakly bound SAMs on ITO with first layer sensitivity across three different SAM systems: [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz), [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz), and (4-(7H-dibenzo[*c,g*]carbazol-7-yl)butyl)phosphonic acid (4PADCB). We employed density functional theory (DFT) calculations and a selective solvent washing strategy to progressively remove SAM molecules based on their binding strength to the ITO substrate, quantifying coverage after each washing step using XPS and cyclic voltammetry (CV). This approach enabled the reconstruction of both the initial distribution and dynamic evolution of SAM adsorption modes on the ITO surface during solvent washing, which included the use of DMF as a model perovskite precursor solution (Figure 1d–f).

Analysis of XPS and CV results first revealed that, contrary to the assumption that SAMs form single molecular layers, all

three SAMs consist of multiple layers. By selectively probing the first layer through CV measurements, however, we found that ethanol washing led to a 15–37% reduction in SAM areal density, which we attribute to the removal of very weakly adsorbed molecules. Meanwhile, DMF washing caused a greater reduction of 31–47%, indicating the additional removal of moderately weakly bound species. These findings not only quantify the abundance of different SAM adsorption modes on ITO but also demonstrate that nearly one-half of the surface-bound molecules are weakly adsorbed and susceptible to removal by DMF, the solvent used to model a perovskite precursor solution.

We then introduced a strategy to mitigate the loss of SAM molecules after the use of DMF through employing a redeposition step after solvent washing of the initial SAM. With this strategy, we achieved a 13–21% increase in relative SAM coverage after DMF washing, indicating that a subset of redeposited SAMs is resistant to solvent-induced removal. Devices fabricated using the redeposition strategy exhibited a 5-fold enhancement in operational stability compared to devices with unwashed SAMs, emphasizing the critical role that SAM stability plays in device longevity.

RESULTS AND DISCUSSION

DFT Calculations of SAM Adsorption to and Removal from ITO. We first aimed to better understand how SAM molecules adsorb to ITO and the relative strength of each adsorption mode. To this end, we performed DFT calculations to evaluate the adsorption energies of 2PACz, MeO-2PACz, and 4PADCB in different adsorption modes. We selected 2PACz as the reference SAM, MeO-2PACz to probe the effect of terminal group substitution, and 4PADCB to investigate the impact of a longer linker group and extended conjugation (Figure S1), enabling the study of a broad range of SAM structures. Moving forward, we will focus primarily on MeO-2PACz in the main text. Results for 2PACz and 4PADCB, which follow the same general trends as MeO-2PACz unless otherwise noted, are provided in the Supporting Information. Furthermore, Table S1 provides definitions of key terms used throughout the manuscript.

For the following DFT calculations, we considered three different adsorption mechanisms that can take place between the SAM molecule and ITO (simulated as the (111) facet of In_2O_3). The first mechanism is a reactive adsorption process (Figure 2a for MeO-2PACz and Figure S2 for 2PACz and 4PADCB) through the heterocondensation reaction explained previously. For this reaction, two indium atoms on the In_2O_3 (111) surface were terminated with hydroxyl groups to simulate a plasma-treated surface.²⁹ The adsorption energy through reactive adsorption was computed after the deprotonation step and formation of water during the heterocondensation reaction to directly quantify the strength of the adsorbed SAM molecule. The other two mechanisms we considered involve adsorption onto nonhydroxylated In_2O_3 (111) sites, where reactive adsorption is not possible. In these scenarios, SAM molecules adsorb nonreactively either through the phosphonic acid anchoring group via hydrogen bonding, where the molecule is oriented perpendicular to the surface so that the phosphonic acid group could hydrogen-bond (Figure 2b for MeO-2PACz and Figure S3 for 2PACz and 4PADCB), or through the carbazole group, where the molecule is oriented parallel to the surface so that the carbazole group could interact with the surface (Figure 2c for MeO-2PACz and Figure S4 for 2PACz and 4PADCB). In both cases, the phosphonic acid of the SAM molecule retains its hydrogen atoms upon adsorption. Because the ITO surface may not be fully hydroxylated, it is plausible that a SAM molecule forms only a single covalent bond. Accordingly, a configuration in which the phosphonic acid forms one covalent bond and one hydrogen bond with its two hydroxyl groups was also modeled (Figure S5).

As shown in Figure 2d, the adsorption energy of MeO-2PACz bound via reactive adsorption was calculated to be -3.16 eV, whereas nonreactive adsorption through the phosphonic acid group (hydrogen bonding) and the carbazole group yielded weaker values of -1.75 and -1.35 eV, respectively. The adsorption energy of MeO-2PACz bound through a single covalent bond was calculated to be -2.29 eV, indicating weaker adsorption than the two-bond configuration but stronger adsorption than either nonreactive adsorption mechanism (Figure S6). Since typical ITO surfaces are only partially hydroxylated and because all adsorption energies are negative, these results indicate that all three adsorption modes—reactive, hydrogen bonding, and carbazole—can coexist, with binding strength decreasing in that order. The

calculated adsorption energies of 2PACz and 4PADCB, which follow the same trend, are shown in Figure S7.

We next sought to investigate whether different adsorption modes within the SAM could be distinguished from each other using a selective removal strategy, in which solvents of varying hydrogen bonding characteristics are used to progressively remove weakly adsorbed molecules according to their binding strength. To achieve this, we calculated the energy required to remove SAM molecules of different adsorption modes using different solvents (termed here as removal energy), where a lower value indicates more favorable removal. Ethanol, DMF, and DMSO were selected as solvents because ethanol is a polar protic solvent with moderate hydrogen bond acceptor and donor ability, while DMF and DMSO are polar aprotic solvents with stronger hydrogen bond acceptor strengths but no hydrogen bond donation ability.^{35,36} We postulated that this contrast would enable us to probe a wide spectrum of adsorption mechanisms between SAM molecules and the ITO surface. These solvents also present practical considerations related to PSC fabrication, mainly that ethanol is widely used as a rinsing solvent during SAM fabrication, while DMF and DMSO are the two solvents in most perovskite precursor formulations.

First, we examined the removal of reactively adsorbed SAM molecules and found that they are stable against ethanol, DMF, and DMSO washing (Figure 2e and Figure S8; final optimized structures shown in Figures S9–S11). For ethanol washing, the large, positive removal energy (2.65 eV for MeO-2PACz) suggests that the desorption process is thermodynamically unfavorable. For DMF and DMSO washing, both solvents are considered to be ineffective at displacing reactively adsorbed SAMs due to their lack of hydrogen bond donation sites.

For hydrogen-bonded MeO-2PACz, ethanol washing leads to a removal energy of 1.32 eV, lower than that of reactively adsorbed MeO-2PACz, indicating that hydrogen-bonded molecules are more likely to be removed than the reactively adsorbed ones. With DMF and DMSO, the removal energies were further reduced to 1.00 and 0.87 eV, respectively, suggesting that DMF and DMSO are more effective than ethanol at removing hydrogen-bonded MeO-2PACz, which is ascribed to their stronger hydrogen-bond-accepting ability.

Lastly, for carbazole-bound MeO-2PACz, the removal energies for ethanol, DMF, and DMSO, were calculated to be 0.92 , 0.61 , and 0.47 eV, respectively, which are the lowest among all modes, indicating that carbazole-bound molecules are the easiest to remove among the three studied modes. 2PACz and 4PADCB follow the same overall trend: reactively adsorbed species remain the most stable across all solvents, hydrogen-bonded species are more easily removed by DMF and DMSO than by ethanol, and carbazole-bound species are the weakest bound and most readily displaced by any solvent. Together, these results suggest that carefully chosen solvents can serve as practical probes to distinguish between different adsorption modes, thereby providing insight into the binding configurations present in SAMs on TCO substrates.

Quantification of SAM Adsorption Modes on ITO.

With DFT calculations identifying selective solvent washing as a potential method to distinguish different SAM adsorption modes, we sought to experimentally validate its effectiveness. As a preface, our computational studies indicate that DMF and DMSO exhibit comparable removal energies for the three SAMs from ITO. Furthermore, experimental results show no significant difference in SAM removal efficacy between DMF,

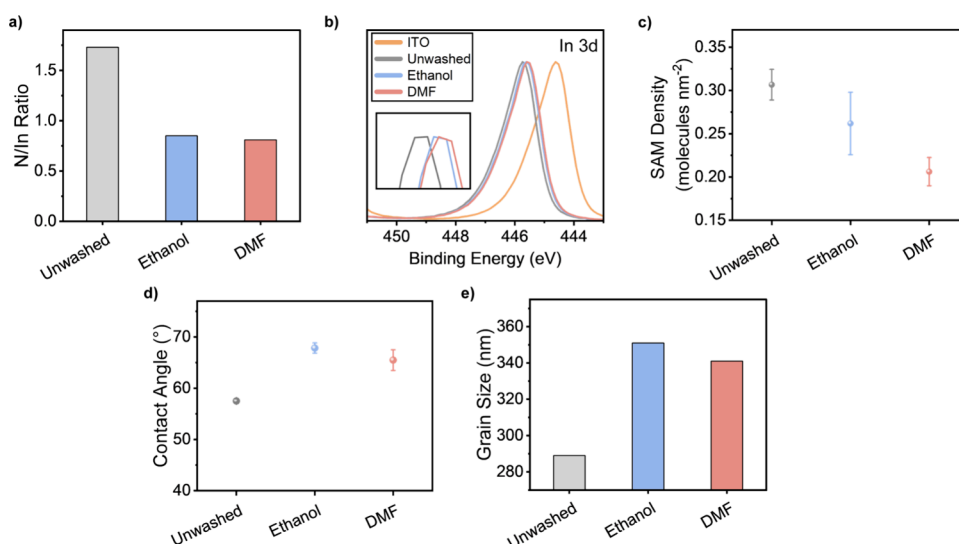


Figure 3. Quantifying the distribution of SAM adsorption modes on ITO. (a) Ratio of nitrogen to indium extracted from XPS spectra of unwashed, twice ethanol-washed, and twice DMF-washed MeO-2PACz. (b) In 3d_{5/2} XPS spectra for bare ITO and corresponding unwashed and solvent-washed MeO-2PACz. Peak shifts are highlighted on the inset. (c) Areal density of unwashed, ethanol-washed, and DMF-washed MeO-2PACz extracted from CV. (d) Average water contact angle of unwashed and solvent-washed MeO-2PACz. (e) Average grain size of perovskite films deposited on unwashed and solvent-washed MeO-2PACz.

DMSO, and the perovskite precursor solution (Figure S12). Therefore, for the following experiments, we selected ethanol and DMF as two representative solvents to selectively wash SAM molecules off ITO substrates. Additionally, we observed negligible changes in the relative SAM abundance from XPS after two consecutive washes with either ethanol or DMF, indicating that all removable molecules had been effectively eliminated after two washing steps (Figure S13, see methods in the SI for detailed SAM processing procedures). Thus, for consistency, all subsequent solvent washing experiments were conducted with two washing steps.

We first sought to determine how the abundance of SAM molecules on ITO evolves across different solvent washing conditions. To achieve this, we first used XPS, analyzing the N 1s spectra after each solvent wash (Figures S14–S16). Since nitrogen is unique to SAMs and indium is unique to ITO, measuring changes in the N/In ratio after solvent washing allowed us to track the resulting change in SAM coverage. XPS results in Figure 3a show that after ethanol washing, the N/In ratio for MeO-2PACz deposited on ITO decreases by 51% (from 1.73 to 0.85). Meanwhile, after DMF washing, the N/In ratio decreases by 55% (from 1.73 to 0.78), indicating that while both ethanol and DMF remove over half of the SAM molecules from ITO, DMF shows a slightly higher removing efficiency. Although 2PACz and 4PADCB show different magnitudes of SAM removal after ethanol and DMF washing, the same trend holds for both SAMs (Figure S17). We further calculated the ratio of phosphorus to indium (with phosphorus also unique to SAMs) and observed a comparable decrease in the N/In ratio for ethanol- and DMF-washed samples (Figures S18–S20), further validating the extent of SAM removal by different solvents.

To further probe the selective removal of SAM molecules from ITO, we again performed XPS, this time focusing on the In 3d_{5/2} core level. Shifts in the In 3d_{5/2} peak reflect changes in the electronic environment of indium atoms within the ITO substrate, allowing the assessment not only of the extent of SAM removal but also the structural evolution of SAMs after

solvent washing. As shown in Figure 3b, upon deposition of MeO-2PACz, we observe a blueshift of +1.12 eV, indicating electron withdrawal from indium atoms due to SAM adsorption and interfacial dipole formation. Ethanol washing then induces a modest redshift of −0.12 eV, consistent with partial removal of noncovalently adsorbed species, while DMF causes a greater redshift of −0.16 eV, suggesting additional loss of weakly bound molecules. While 2PACz exhibits the same behavior as MeO-2PACz (Figure S21), 4PADCB shows distinct peak-shifting trends that reveal its unique structural evolution during solvent washing (Figure S22), which are detailed in Supplementary Note 1. These results highlight how indium XPS serves as a powerful tool for capturing not only the extent of SAM removal but also the unique structural reconfigurations that different SAM chemistries undergo during solvent washing.

While XPS is valuable for tracking the overall abundance of SAM molecules, it cannot distinguish between molecules that are adsorbed directly on the ITO surface and those that are potentially loosely packed in upper layers. This limitation arises from the large penetration depth (50–100 Å) of XPS,³⁷ resulting in signal contributions from all layers of the SAM structure. As a result, XPS alone cannot directly isolate changes in the population of first-layer SAM molecules after solvent washing, a parameter more relevant to PSC performance.

To overcome these limitations and selectively probe the removal of first-layer SAMs, we turned to CV, an electrochemical technique that is inherently first-layer sensitive; that is, it quantifies the areal density of electrochemically active molecules, such as carbazole-based SAMs, that are directly coupled to the electrode surface (Figures S23–S31). Unlike XPS, CV selectively probes molecules that are either (i) directly adsorbed onto the ITO surface³⁸ or (ii) positioned close enough to ITO to allow efficient electron tunneling, typically within 1.5 nm of the surface.³⁹ Given that the molecular length of adsorbed SAM molecules is slightly below 1.5 nm (Figures S32–S34), molecules in the second layer and

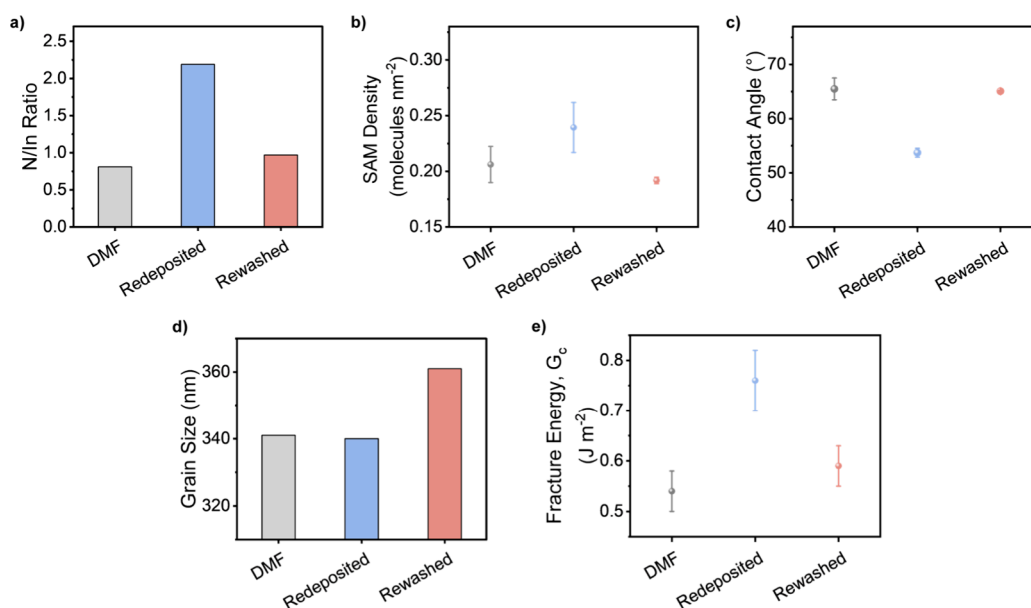


Figure 4. Redeposition strategy to reduce the solvent-induced removal of SAM molecules from the ITO surface. (a) N/In ratio extracted from XPS spectra of DMF-washed, redeposited, and rewashed MeO-2PACz films. (b) Areal density of DMF-washed, redeposited, and rewashed MeO-2PACz films extracted from CV. (c) Average water contact angle of DMF-washed, redeposited, and rewashed MeO-2PACz. (d) Average grain size of perovskite films deposited on DMF-washed, redeposited, and rewashed MeO-2PACz. (e) Fracture energy values of the MeO-2PACz/perovskite interface in DMF-washed, redeposited, and rewashed samples.

beyond are likely positioned further from the surface than the tunneling limit and thus do not contribute to the CV signal.

CV results shown in Figure 3c demonstrate that ethanol washing reduces MeO-2PACz areal density from 0.307 to 0.262 molecules nm⁻² (a 15% reduction), while DMF washing leads to a more pronounced decrease to 0.206 molecules nm⁻² (a 33% reduction). We first note that while XPS shows nearly identical overall decreases in SAM abundance for ethanol and DMF treatments (a 3% difference in the N/In ratio), CV reveals a much larger difference between the two (a 10% difference in areal density). Given that both solvents are able to remove loosely bound molecules in secondary layers, this difference suggests the presence of a multilayered SAM structure, with a substantial fraction of molecules extending beyond the first monolayer. Additionally, the stronger reduction in CV-derived areal density after DMF washing (33%) indicates that DMF is far more effective than ethanol (15%) at removing weakly bound molecules directly adsorbed to the ITO surface. For 2PACz and 4PADCB (Figure S35), the overall trends mirror those observed for MeO-2PACz.

These combined XPS and CV results suggest that ethanol removes a relatively small fraction of SAM molecules (15–37%), corresponding to the very weakly bound molecules such as carbazole-bound species. In contrast, DMF removes 31–47% of the SAM molecules by stripping away not only these very weakly bound molecules but also moderately bound ones such as those stabilized by hydrogen bonding. We further conclude that the first-layer SAM molecules that remain intact after DMF washing (53–69%) correspond to those that are covalently bound to the ITO surface.

Impact of SAM Removal on Perovskite Film Formation. Next, we investigated how changes in SAM coverage influence perovskite film formation. Since perovskite is deposited directly onto the SAM-covered ITO, variations in coverage and adsorption mode distribution will affect surface wettability and, in turn, modulate the perovskite growth

process.^{40,41} To evaluate the wettability of the perovskite precursor solution on the SAM-covered ITO, we examined the water contact angle of the SAM-covered ITO surface before and after different washing treatments (Figure S36 and Table S2). As shown in Figure 3d, ethanol and DMF washing increased the average water contact angle on MeO-2PACz from 57.5° in the unwashed sample to 67.8° and 65.5°, respectively, indicating a more hydrophobic surface after solvent exposure that would hinder the uniform wetting of the perovskite precursor solution. This change suggests that loosely bound molecules from the upper layers, particularly those oriented with their hydrophilic phosphonic acid groups exposed, are removed, leaving behind more hydrophobic functional groups at the SAM surface. 2PACz and 4PADCB show the same trend: contact angles increase after solvent washing, consistent with the removal of weakly bound, hydrophilic SAM molecules and the exposure of more hydrophobic regions at the surface (Figures S37–S39 and Tables S3 and S4).

To investigate how SAM removal influences perovskite morphology, we deposited Cs_{0.06}FA_{0.94}PbI₃ perovskite thin films onto ITO/SAM substrates under unwashed and solvent-washed conditions (see methods for perovskite deposition procedures) and took scanning electron microscopy (SEM) images to determine how grain size evolves across washing conditions. SEM data shown in Figure 3e and Figure S40 show that perovskite films deposited on unwashed MeO-2PACz exhibited an average grain size of 289 nm, which increased to 351 nm after ethanol washing and 341 nm after DMF washing. This grain size enlargement is attributed to the removal of loosely bound, disordered SAM molecules, which improves the structural uniformity of the SAM and reduces interfacial disorder, thereby facilitating more uniform nucleation and coherent perovskite crystal growth.⁴² 2PACz and 4PADCB follow the same trend, where grain sizes increase after ethanol and DMF washing, consistent with the idea that solvent-

induced removal of disordered upper-layer SAM molecules enhances SAM uniformity and promotes improved perovskite crystallization (Figure S41).

Redeposition of SAMs on ITO Substrates. We have shown that exposure of SAMs to a perovskite precursor solvent (i.e., DMF) results in the removal of over half of the total molecules from the surface, including a substantial loss of those within the first layer. This depletion can impair device performance by lowering the ITO work function,⁴³ which disrupts band alignment with the perovskite and hinders hole extraction, while also increasing the likelihood of shunting and reducing device reproducibility.⁴⁴ Therefore, to maximize PSC performance, it is critical to develop strategies that improve the interfacial stability of SAMs, ensuring their resistance to solvent-induced removal.

To increase the number of SAM molecules on the ITO surface that are resistant to solvent-induced removal, we implemented a redeposition strategy (see methods for detailed protocols). Briefly, SAM molecules were redeposited onto a substrate that had already been coated with SAMs previously and washed with DMF, yielding “redeposited” samples. This approach was motivated by the hypothesis that removing weakly bound, disordered SAM molecules during the initial DMF wash may reduce steric congestion within the SAM structure, potentially creating a more favorable environment for subsequent molecular adsorption.

To quantify changes in molecular abundance and ITO surface coverage after redeposition, we performed XPS and CV measurements and compared them to the DMF-washed samples. XPS analysis (Figure 4a) revealed that the N/In ratio of MeO-2PACz increased from 0.78 to 2.41, indicating a substantial addition of SAM molecules on the ITO surface. In contrast, CV (Figure 4b and Figure S42) showed a more modest increase in areal density, from 0.206 to 0.239 molecules nm⁻². The difference between the large increase in XPS signal and the smaller CV change suggests that while some redeposited molecules directly adsorb to vacant ITO sites, the majority accumulate above the primary monolayer as secondary layers. 2PACz and 4PADCB show the same behavior, with redeposition increasing the total number of SAM molecules detected by XPS more than the areal density measured by CV, consistent with the formation of loosely bound secondary overlayers (Figures S43–S46).

Assessing the Stability and Orientation of Redeposited SAMs. To assess the stability of redeposited SAM molecules, we subjected redeposited samples to a second DMF washing step, yielding “rewashed” samples. As shown in Figure 4a and Figure S43, XPS measurements revealed that N/In ratios of rewashed samples remained higher than those of the DMF-washed baseline (i.e., samples that underwent one DMF washing step) by 13%, 20%, and 21% for MeO-2PACz, 2PACz, and 4PADCB, respectively, demonstrating that redeposition introduces more SAM molecules onto the ITO surface that are stable against DMF washing. Interestingly, for all three SAMs, the rewashed samples exhibit nearly identical areal densities (derived from CV measurement) compared to the DMF-washed baseline samples (Figures S47–S49). Together, these findings indicate that the stable, redeposited molecules detected by XPS are likely located above the first monolayer (i.e., not covalently bound to the substrate) and therefore do not contribute to the CV signal. Analysis of the In 3d_{5/2} peak further confirmed that these redeposited molecules, although

resistant to DMF washing, are not covalently bound to the ITO substrate (Figure S50 and Supplementary Note 2).

To better understand how these stable, redeposited SAM molecules are arranged on the ITO substrate, we conducted water contact angle measurements (Figure S51 and Table S5). As shown in Figure 4c, redeposition of MeO-2PACz led to a decrease in the average contact angle from 65.5° (initial DMF wash) to 53.7°. This increased hydrophilicity suggests that polar phosphonic acid groups are exposed at the surface of the secondary layers. Upon rewashing, however, the contact angle returns to 65.1°, closely matching that of the initial DMF-washed SAM. This recovery, coupled with the previous CV and In 3d_{5/2} XPS data, suggests that redeposited, DMF-stable MeO-2PACz molecules align in the same upright orientation as the initial chemisorbed layer, but positioned slightly above the surface to avoid direct adsorption to the underlying ITO. 2PACz and 4PADCB follow the same trend as MeO-2PACz (Figures S52–S54 and Tables S6 and S7).

Given that the redeposited, DMF-stable SAM molecules and already chemisorbed SAM molecules are oriented in the same direction, we posited that intermolecular interactions could be a potential mechanism that induces the observed DMF stability. To probe these potential interactions, we analyzed the N 1s peak of the carbazole nitrogen in MeO-2PACz, whose electron density can reflect intermolecular interactions such as π – π stacking (Figures S55 and S56). Shifts in peak position and reductions in full width at half-maximum upon redeposition and rewashing were observed across all three SAMs, consistent with increased intermolecular interactions and suggesting this as a plausible mechanism for the observed DMF stability.

To further evaluate how SAM redeposition and subsequent rewashing influence molecular orientation at the interface and, in turn, perovskite crystallization, we performed SEM analysis to quantify grain size evolution across treatments (Figure 4d and Figure S57). For MeO-2PACz, the perovskite grain size remained essentially unchanged after redeposition (341 nm after DMF washing vs 340 nm), suggesting that although additional molecules adsorb during redeposition, their partially disordered orientation does not substantially alter the perovskite nucleation landscape. After rewashing with DMF, however, the grain size increased to 361 nm, consistent with removal of loosely bound molecules and retention of an upright SAM configuration that promotes uniform nucleation and growth. 2PACz followed a similar trend to MeO-2PACz, reflecting their structural similarity. In contrast, for 4PADCB, grain size increased from 345 to 356 nm upon redeposition but decreased to 336 nm after rewashing (Figure S58). This trend aligns with its flat-lying orientation after DMF washing, indicating that redeposition promotes further horizontal stacking rather than upright alignment, which hinders perovskite crystallization.

To probe how SAM redeposition influences the SAM/perovskite interaction, we conducted fracture energy (G_c) measurements using the sandwich double-cantilever beam delamination method.^{45,46} Since the mechanical failure occurs at the weakest interface being studied, which in this case is the SAM/perovskite interface,^{47,48} comparing G_c values across different conditions provides direct insight into how interfacial interactions evolve. As shown in Figure 4e, DMF-washed samples exhibit an average G_c of 0.54 J m⁻². After redeposition, this value increases to 0.76 J m⁻². The improvement in G_c after redeposition can be attributed to a combination of improved

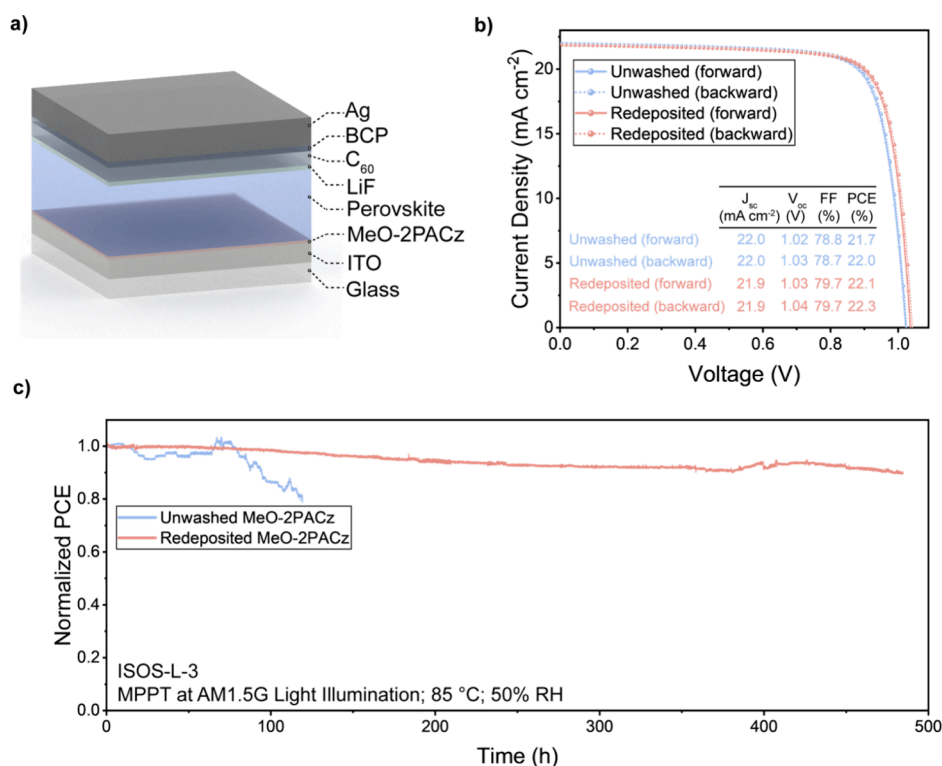


Figure 5. Impact of SAM redeposition on PSC performance. (a) Illustration of the PSC device architecture. (b) J - V curves of the best-performing unwashed and redeposited MeO-2PACz-based device. (c) Operational stability of encapsulated MeO-2PACz-based PSCs under constant 1-sun illumination at 85 °C in air with 50% relative humidity.

surface coverage and improved SAM ordering and orientation (Figure S59 and Supplementary Note 3).⁴⁹ After rewashing, G_c decreases to 0.59 J m⁻², reflecting the removal of most redeposited molecules, reducing surface density and thereby lowering interfacial toughness.

The combined results from this section indicate that redeposition enhances SAM stability by introducing additional molecules that, although not chemisorbed themselves, are likely stabilized through intermolecular interactions with the already chemisorbed layer. This stabilizing mechanism produces a SAM structure that resists solvent removal, promotes improved perovskite crystallization, and strengthens adhesion at the SAM/perovskite interface.

Impact of SAM Redeposition on PSC Performance. To assess the impact of SAM redeposition on PSC performance, we fabricated devices with the architecture ITO/MeO-2PACz/Perovskite/LiF/C₆₀/BCP/Ag (Figure 5a). As shown in Figure 5b, the control devices with unwashed MeO-2PACz achieved a maximum power conversion efficiency (PCE) of 21.7% (22.0% under reverse scanning), which increased to 22.1% (22.3% under reverse scanning) after redeposition—the highest PCE observed among all processing conditions tested (Figure S60 and Table S8). We then tested the stability of devices with redeposited MeO-2PACz by subjecting encapsulated devices to ISOS-L-3 protocols (maximum power point tracking (MPPT) at 85 °C and 50% relative humidity (RH)), measuring their efficiency over time. As shown in Figure 5c, control devices with unwashed MeO-2PACz rapidly degraded to 90% of their initial efficiency after 94 h. In contrast, devices with redeposited MeO-2PACz retained 90% of their initial efficiency after 480 h, a 5-fold improvement over the unwashed control device. We ascribe the enhanced operational stability

to a denser, more stable SAM layer formed after redeposition, which strengthens SAM–perovskite adhesion, as evidenced by the G_c results shown in Figure 4e.⁵⁰

CONCLUSIONS

In conclusion, we developed the first integrated computational and experimental framework to quantify the distribution of 2PACz, MeO-2PACz, and 4PADCB adsorption modes within the first layer of a SAM structure using a simulated perovskite deposition process. DFT calculations suggested that ethanol and DMF can selectively remove different adsorption modes, with DMF used here as a model perovskite precursor solvent, capable of displacing a broader range of adsorption modes. XPS was then employed to measure the total molecular abundance across multilayered SAMs, while CV was used to selectively measure the density of molecules in the first monolayer. This dual characterization revealed that DMF washing removes over 50% of SAMs, including a substantial reduction in the first layer, through displacement of weakly bound species. To mitigate these losses, we introduced a redeposition strategy, which improved SAM retention by 20% (2PACz), 13% (MeO-2PACz), and 21% (4PADCB) after DMF exposure. Notably, modulating SAM coverage via solvent washing and redeposition significantly impacted PSC performance metrics, with redeposition leading to a 5-fold improvement in operational stability. Overall, this study establishes a robust framework for tracking SAM evolution throughout PSC fabrication and offers a simple yet effective method for improving PSC performance under real-world processing conditions.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c15955>.

Detailed methods for experimental procedures, DFT calculations, characterizations, XPS spectra, CV curves and fitted data, contact angle images, SEM images, and device performance statistics for all SAM processing conditions (PDF)

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Notes

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